

Supplementary tables from the evaluation results

The results of the evaluation of the developed mechanism for Cuboid Splatting-with-prior are presented in the tables below. The evaluation was based on the following research questions:

Main question: Inhowfar does the correction of a prior scenario work?

Sensitivity analyses:

1. How well does the translation of a prior scenario into a cuboid prior work?
2. What happens when a (nearly) perfect prior scenario is used?
3. What happens when a completely wrong prior scenario is used?
4. Can the method correct source term errors from the prior scenario?
5. Can the method correct weather data errors from the prior scenario?
- (6. Can the method correct incorrectly (un)polluted parts of the prior scenario?)

For each of these questions, the experiments were evaluated using matching prior scenarios made available from the RODOS system or actively constructed for evaluation purposes. The results are presented in the tables below.

Inhowfar does the correction of a prior scenario work?

The following priors are used for evaluation, which are fundamentally suitable in terms of their prior quality (scenario error below 100%, roughly useful shape):

Emsland:

- EuRim Prior
- EuRim prior x0,8
- Emsland area-wise medium random noise

Brokdorf:

- Brokdorf_Prior1
- Brokdorf_Prior3
- Brokdorf_Prior1 x0,8
- Brokdorf area-wise medium random noise

The results regarding the reconstruction qualities achieved and the baseline comparisons regarding prior quality and reconstruction quality with Cuboid Splatting without prior are presented in Tables 1 and 2. These take into account the experiments with 25,000 paths in all three resolutions (30x30, 10x10, 100x100) using the ADC mechanism. The metrics refer to relevance 1.

Table 1: Reconstruction results for the Emsland scenario with cuboid splatting using a prior derived from suitable prior scenarios from 25,000 paths. All metrics regarding relevance 1.

Prior	Resolu- tion	Reconstruc- tion quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
EuRim Prior	30x30	33,38%	80,37%	(+46,99%P)	40,14%	(+6,75%P)
EuRim Prior x0.8	30x30	34,42%	84,3%	(+49,87%P)	40,14%	(+5,72%P)
Emsland Noise medium	30x30	13,13%	20,27%	(+7,14%P)	40,14%	(+27%P)

EuRim Prior	10x10	9,42%	79,81%	(+70,4%P)	17,74%	(+8,33%P)
EuRim Prior x0.8	10x10	10,79%	83,85%	(+73,06%P)	17,74%	(+6,96%P)
Emsland Noise medium	10x10	5,88%	22,21%	(+16,33%P)	17,74%	(+11,86%P)
EuRim Prior	100x100	59,85%	91,95%	(+32,1%P)	145,31%	(+85,46%P)
EuRim Prior x0.8	100x100	59,95%	93,48%	(+33,53%P)	145,31%	(+85,36%P)
Emsland Noise medium	100x100	18,54%	19,82%	(+1,28%P)	145,31%	(+126,77%P)

Table 2: Reconstruction results for the Brokdorf scenario with cuboid splatting using a prior derived from suitable prior scenarios from 25,000 paths. All metrics regarding relevance 1.

Prior	Resolution	Reconstruction quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
Brokdorf Prior1	30x30	8,91%	24,12%	(+15,21%P)	122,25	(+113,33%P)
Brokdorf Prior3	30x30	12,04%	38,63%	(+26,58%P)	122,25	(+110,2%P)
Brokdorf Prior1 x0.8	30x30	13,81%	33,2%	(+19,4%P)	122,25	(+108,44%P)
Brokdorf Noise medium	30x30	12,88%	31,46%	(+18,58%P)	122,25	(+109,37%P)
Brokdorf Prior1	10x10	8,79%	19,14%	(+10,35%P)	47,87	(+39,08%P)
Brokdorf Prior3	10x10	11,6%	32,11%	(+20,51%P)	47,87	(+36,27%P)
Brokdorf Prior1 x0.8	10x10	8,94%	31,64%	(+22,7%P)	47,87	(+38,93%P)
Brokdorf Noise medium	10x10	8,94%	30,07%	(+21,14%P)	47,87	(+38,93%P)
Brokdorf Prior1	100x100	20,01%	47,06%	(+27,05%P)	252,21	(+232,2%P)
Brokdorf Prior3	100x100	24,1%	62,31%	(+38,21%P)	252,21	(+228,11%P)
Brokdorf Prior1 x0.8	100x100	19,73%	42,46%	(+22,74%P)	252,21	(+232,48%P)
Brokdorf Noise medium	100x100	16,71%	31,94%	(+15,23%P)	252,21	(+235,5%P)

How well does the translation of a prior scenario into a cuboid prior work?

Tables 3, 4, and 5 show a set of aggregated metrics for translating prior scenarios into cuboid priors in the final evaluation.

Table 3: Aggregated metrics for translating prior scenarios to cuboid priors for the final evaluation in 30x30 resolution

	Emsland priors		Brokdorf priors	
Metric	Mean	Min - Max	Mean	Min - Max
Translation error [%]	1,02 %	0,18 - 2,65 %	1,26 %	0,43 - 3,78 %
Number of Cuboids	631	616 - 640	616	466 - 633
Duration [Min]	14 Min	10 - 16 Min	7 Min	6 - 8 Min
Error at Init. [%]	635,52 %	390,59 - 951,32 %	733,49 %	214,32 - 2153,19 %
Information loss [%P]	-0,9 %P	-5,17 - 0,01 %P	- 1,58 %P	-19,09 %P - 0,01 %P

Table 4: Aggregated metrics for translating prior scenarios to cuboid priors for the final evaluation in 10x10 resolution

	Emsland priors		Brokdorf priors	
Metric	Mean	Min - Max	Mean	Min - Max
Translation error [%]	0,27 %	0,05 - 0,69 %	0,44 %	0,15 - 1,02 %
Number of Cuboids	561	557 - 565	551	481 - 563
Duration [Min]	2 Min	1 - 2 Min	1 Min	0,5 - 1 Min
Error at Init. [%]	573,86 %	270,59 - 775,23 %	654,29 %	322,07 - 912,14 %
Information loss [%P]	-0,43 %P	-3,16 - 0,0 %P	-0,37 %P	-2,74 - 0,0 %P

Table 5: Aggregated metrics for translating prior scenarios to cuboid priors for the final evaluation in 100x100 resolution

	Emsland priors		Brokdorf priors	
Metric	Mean	Min - Max	Mean	Min - Max
Translation error [%]	3,98 %	0,85 - 7,31 %	2,86 %	1,5 - 4,61 %
Number of Cuboids	599	533 - 615	627	610 - 638
Duration [Min]	116 Min	94 - 154 Min	73 Min	56 - 89 Min
Error at Init. [%]	1593,32 %	453,48 - 3813 %	986,38 %	184,07 - 2820,7 %
Information loss [%P]	-2,26 %P	-6,69 - 0,02 %P	-1,41 %P	-4,81 - 0,07 %P

What happens when a (nearly) perfect prior scenario is used?

The following priors are used for evaluation, which are perfect (0% scenario error) or nearly perfect (2 to 3% scenario error) in terms of their prior quality:

Emsland:

- Emsland area-wise small random noise
- Emsland perfect

Brokdorf:

- Brokdorf area-wise small random noise
- Brokdorf perfect

The results of these experiments with Cuboid Splatting with prior are presented in Tables 6 and 7.

Table 6: Reconstruction results for the Emsland scenario with cuboid splatting using prior from (nearly) perfect prior scenarios from 25,000 paths with ADC. Metrics regarding relevance 1.

Prior	Resolution	Reconstruction quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
Emsland Noise small	30x30	11,71%	2,03%	(-9,69%P)	40,14%	(+28,42%P)
Emsland perfect	30x30	7,86%	0%	(-7,86%P)	40,14%	(+32,28%P)
Emsland Noise small	10x10	3,29%	2,22%	(-1,07%P)	17,74%	(+14,45%P)
Emsland perfect	10x10	2,66%	0%	(-2,66%P)	17,74%	(+15,08%P)
Emsland Noise small	100x100	18,84%	1,98%	(-16,86%P)	145,31%	(+126,47%P)
Emsland perfect	100x100	12,83%	0%	(-12,83%P)	145,31%	(+132,48%P)

Table 7: Reconstruction results for the Brokdorf scenario with cuboid splatting using prior from (nearly) perfect prior scenarios from 25,000 paths with ADC. Metrics regarding relevance 1.

Prior	Resolution	Reconstruction quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
Brokdorf Noise small	30x30	3,22%	3,15%	(-0,08%P)	122,25%	(+119,02%P)
Brokdorf perfect	30x30	4,86%	0%	(-4,86%P)	122,25%	(+117,38%P)
Brokdorf Noise small	10x10	1,05%	3,01%	(+1,96%P)	47,87%	(+46,82%P)
Brokdorf perfect	10x10	0,58%	0%	(-0,58%P)	47,87%	(+47,28%P)
Brokdorf Noise small	100x100	12,83%	3,19%	(-9,64%P)	252,21%	(+239,38%P)
Brokdorf perfect	100x100	8,66%	0%	(-8,66%P)	252,21%	(+243,55%P)

What happens when a completely wrong prior scenario is used?

The following priors are used for evaluation, which are completely wrong in terms of their origin or prior quality:

Emsland:

- Brokdorf instead of Emsland (completely wrong)
- Emsland area-wise strong random noise

Brokdorf:

- Emsland instead of Brokdorf (completely wrong)
- Brokdorf area-wise strong random noise

The evaluation of the effects of a completely wrong prior is carried out here on the basis of two constructed prior scenarios: firstly, a swap between the Emsland and Brokdorf scenarios as the prior for the reconstruction of the other original scenario, and secondly, an original scenario with approximately 100% scenario error, with strong area-wise random noise. The following distinction is important for the swapped prior scenarios: if the Brokdorf scenario is used as the prior for the reconstruction of the Emsland scenario, this results in a prior with very high exposure values and thus a scenario error of around 2000%. Conversely, if the Emsland scenario is used as the prior for the reconstruction of the Brokdorf scenario, this results in a prior with load values that are far too low and thus a scenario error of slightly less than 100%.

The results of these experiments with Cuboid Splatting with prior are presented in Tables 8 and 9.

Table 8: Reconstruction results for the Emsland scenario with cuboid splatting with prior from completely wrong prior scenarios from 25,000 paths with ADC. Metrics regarding relevance 1.

Prior	Resolu- tion	Reconstruc- tion quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
Emsland wrong scenario	30x30	63,68%	2044,68%	(+1980,82%P)	40,14%	(-23,72%P)
Emsland Noise large	30x30	23,12%	105,96%	(+82,84%P)	40,14%	(+17,02%P)
Emsland wrong scenario	10x10	22,19%	1785,44%	(+1763,25%P)	17,74%	(-4,45%P)
Emsland Noise large	10x10	9%	124,09%	(+115,09%P)	17,74%	(+8,75%P)
Emsland wrong scenario	100x100	139,5%	1639,47%	(+1499,96%P)	145,31%	(+5,8%P)
Emsland Noise large	100x100	38,53%	96,43%	(+57,9%P)	145,31%	(+106,77%P)

Table 9: Reconstruction results for the Brokdorf scenario with cuboid splatting with prior from completely wrong prior scenarios from 25,000 paths with ADC. Metrics regarding relevance 1.

Prior	Resolu- tion	Reconstruc- tion quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	

Brokdorf wrong scenario	30x30	89,2%	96,06%	(+6,86%P)	122,25%	(+33,04%P)
Brokdorf Noise large	30x30	8,16%	97,1%	(+88,94%P)	122,25%	(+114,08%P)
Brokdorf wrong scenario	10x10	32,56%	94,93%	(+62,38%P)	47,87%	(+15,31%P)
Brokdorf Noise large	10x10	7,82%	112,1%	(+104,28%P)	47,87%	(+40,05%P)
Brokdorf wrong scenario	100x100	165,54%	95,43%	(-70,11%P)	252,21%	(+86,67%P)
Brokdorf Noise large	100x100	29,33%	133,14%	(+103,8%P)	252,21%	(+222,88%P)

Can the method correct source term errors from the prior scenario?

The following priors are used for evaluation, which contain the four different types of source term errors:

Emsland:

- Emsland overestimation x2
- Emsland overestimation x10
- Emsland underestimation x0,5
- Emsland underestimation x0,1
- Emsland time-dependent source term error (x2, x0,5)
- Emsland time-dependent source term error (x10, x0,1)

Brokdorf:

- Brokdorf overestimation x2
- Brokdorf overestimation x10
- Brokdorf underestimation x0,5
- Brokdorf underestimation x0,1
- Brokdorf time-dependent source term error (x2, x0,5)
- Brokdorf time-dependent source term error (x10, x0,1)
- Brokdorf_Prior2 (z -5h)

Systematic overestimation: The reconstruction qualities achieved are shown in Table 10, and a more detailed analysis of the correction of the iodine concentration values is shown in Table 11.

Table 10: Reconstruction results for Emsland and Brokdorf using Cuboid Splatting with prior from prior scenarios with systematically overestimated source term. Metrics regarding relevance 1.

Prior	Resolu- tion	Reconstruc- tion quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
Emsland x2	30x30	15,86%	100,00%	(+84,14%P)	40,14%	(+24,28%P)
Emsland x10	30x30	57,98%	900,00%	(+842,02%P)	40,14%	(-17,84%P)
Emsland x2	10x10	8,80%	100,00%	(+91,20%P)	17,74%	(+8,95%P)
Emsland x10	10x10	25,70%	900,00%	(+874,30%P)	17,74%	(-7,96%P)

Emsland x2	100x100	34,33%	100,00%	(+65,67%P)	145,31%	(+110,98%P)
Emsland x10	100x100	158,03%	900,00%	(+741,97%P)	145,31%	(-12,72%P)
Brokdorf x2	30x30	9,42%	100,00%	(+90,58%P)	122,25%	(+112,83%P)
Brokdorf x10	30x30	32,03%	900,00%	(+867,97%P)	122,25%	(+90,21%P)
Brokdorf x2	10x10	4,87%	100,00%	(+95,13%P)	47,87%	(+43,00%P)
Brokdorf x10	10x10	13,43%	900,00%	(+886,57%P)	47,87%	(+34,44%P)
Brokdorf x2	100x100	15,46%	100,00%	(+84,54%P)	252,21%	(+236,75%P)
Brokdorf x10	100x100	33,55%	900,00%	(+866,45%P)	252,21%	(+218,66%P)

Table 11: Analysis of the change in iodine load values based on the overestimated prior scenario through correction with Cuboid Splatting with prior. (30x30, 25,000 paths, with ADC)

	Sum Iodine Values (Relevance 0)		Factor	Sum Iodine Values (Relevance 1)		Prior
	Prior	Rekonstruk- tion		Prior	Rekonstruk- tion	
Emsland x2 (Aim 50%)	4167992,5	2389865,5	57,45%	1521345,1	746337,4	49,06%
Emsland x10 (Aim 10%)	20839964,0	7676178,5	36,83%	7606725,5	995581,4	13,09%
Brokdorf x2 (Aim 50%)	72908730,0	37434000,0	51,34%	60715296,0	30736680,0	50,62%
Brokdorf x10 (Aim 10%)	364543650,0	51720756,0	14,19%	303576480,0	32057332,0	10,56%

Systematic underestimation: The reconstruction qualities achieved are shown in Table 12, and a more detailed analysis of the correction of the iodine concentration values is shown in Table 13.

Table 12: Reconstruction results for Emsland and Brokdorf using Cuboid Splatting with prior from prior scenarios with systematically overestimated source term. Metrics regarding relevance 1.

Prior	Resolu- tion	Reconstruc- tion quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
Emsland x0,5	30x30	24,25%	50,00%	(+25,75%P)	40,14%	(+15,89%P)

Emsland x0,1	30x30	36,85%	90,00%	(+53,15%P)	40,14%	(+3,29%P)
Emsland x0,5	10x10	9,33%	50,00%	(+40,67%P)	17,74%	(+8,41%P)
Emsland x0,1	10x10	13,37%	90,00%	(+76,63%P)	17,74%	(+4,38%P)
Emsland x0,5	100x100	36,00%	50,00%	(+14,00%P)	145,31%	(+109,30%P)
Emsland x0,1	100x100	58,68%	90,00%	(+31,32%P)	145,31%	(+86,63%P)
Brokdorf x0,5	30x30	17,55%	50,00%	(+32,45%P)	122,25%	(+104,70%P)
Brokdorf x0,1	30x30	61,11%	90,00%	(+28,89%P)	122,25%	(+61,14%P)
Brokdorf x0,5	10x10	7,91%	50,00%	(+42,09%P)	47,87%	(+39,96%P)
Brokdorf x0,1	10x10	33,50%	90,00%	(+56,50%P)	47,87%	(+14,37%P)
Brokdorf x0,5	100x100	25,31%	50,00%	(+24,69%P)	252,21%	(+226,90%P)
Brokdorf x0,1	100x100	110,06%	90,00%	(-20,06%P)	252,21%	(+142,15%P)

Table 13: Analysis of the change in iodine load values based on the underestimated prior scenario through correction with Cuboid Splatting with prior. (30x30, 25,000 paths, with ADC)

	Sum Iodine Values (Relevance 0)				Sum Iodine Values (Relevance 1)		
	Prior	Rekonstruk- tion	Factor		Prior	Rekonstruk- tion	Factor
Emsland x0,5 (Aim 200%)	1041998,1	1767686,8	169,64 %		380336,3	765344,9	201,23 %
Emsland x0,1 (Aim 1000%)	208399,6	1783314,9	855,72 %		76067,3	775797,5	1019,8 8%
Brokdorf x0,5 (Aim 200%)	18227182,0	39159120,0	214,84 %		15178824,0	31028554,0	204,42 %
Brokdorf x0,1 (Aim 1000%)	3645436,0	47354532,0	1299,0 1%		3035765,0	34905136,0	1149,8 0%

Time-dependent over- and underestimation: The reconstruction qualities achieved are shown in Table 14, and a more detailed analysis of the correction of the iodine concentration values is shown in Table 15.

Table 14: Reconstruction results for Emsland and Brokdorf using cuboid splatting with prior from priors with time-step-dependent overestimated/underestimated source term. Metrics regarding relevance 1.

Prior	Resolution	Reconstruction quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
Emsland x2 x0,5	30x30	17,21%	22,39%	(+5,18%P)	40,14%	(+22,93%P)
Emsland x10 x0,1	30x30	36,28%	161,04%	(+124,76%P)	40,14%	(+3,85%P)
Emsland x2 x0,5	10x10	8,72%	21,47%	(+12,75%P)	17,74%	(+9,03%P)
Emsland x10 x0,1	10x10	14,34%	142,09%	(+127,75%P)	17,74%	(+3,41%P)
Emsland x2 x0,5	100x100	23,14%	23,53%	(+0,39%P)	145,31%	(+122,16%P)
Emsland x10 x0,1	100x100	98,98%	174,44%	(+75,46%P)	145,31%	(+46,32%P)
Brokdorf x2 x0,5	30x30	5,77%	17,95%	(+12,18%P)	122,25%	(+116,48%P)
Brokdorf x10 x0,1	30x30	9,56%	155,13%	(+145,58%P)	122,25%	(+112,69%P)
Brokdorf x2 x0,5	10x10	5,94%	18,10%	(+12,15%P)	47,87%	(+41,92%P)
Brokdorf x10 x0,1	10x10	8,55%	156,69%	(+148,14%P)	47,87%	(+39,32%P)
Brokdorf x2 x0,5	100x100	20,05%	15,64%	(-4,41%P)	252,21%	(+232,16%P)
Brokdorf x10 x0,1	100x100	22,28%	128,58%	(+106,30%P)	252,21%	(+229,93%P)

*Table 15: Analysis of the change in load values based on the time-dependent overestimated/underestimated prior through correction with cuboid splatting.
(30x30, 25,000 paths, with ADC)*

	Sum Iodine Values (Relevance 0)		Factor	Sum Iodine Values (Relevance 1)		Prior
	Prior	Rekonstruk- tion		Prior	Rekonstruk- tion	
Emsland x2 x0,5 x1 (Aim 50%, 200%, 100%)	584589,9	341465,7	58,41%	255132,6	141437,6	55,44%
	153299,4	180628,0	117,82 %	42736,5	62344,3	145,88 %
	1485102,5	1348613,5	90,81%	547633,3	556536,6	101,63 %

Emsland x10 x0,1 x1 (Aim 10%, 1000%, 100%)	2922949,2	773977,4	26,48%	1275663,0	176843,0	13,86%
	30659,9	1388778,0	452,96 %	8547,3	55071,8	644,32 %
	1485102,5	1454660,2	97,96%	547633,3	528605,0	96,53%
Brokdorf x2 x0,5 x1 (Aim 50%, 200%, 100%)	12237746,0	6620060,0	54,10%	10357838,0	5414860,0	52,28%
	308924,7	524663,7	169,84 %	269243,8	453662,4	168,50 %
	29717642,0	29491926,0	99,24%	24640242,0	24550440,0	99,64%
Brokdorf x 10 x0,1 x1 (Aim 10%, 1000%, 100%)	61188724,0	7567255,0	12,37%	51789190,0	5616957,0	10,85%
	61784,9	522169,3	845,14 %	53848,8	420351,2	780,61 %
	29717642,0	29697266,0	99,93%	24640242,0	24531418,0	99,56%

Systematic error along the time axis: The reconstruction qualities are shown in Table 16.

Table 16: Reconstruction results for the Brokdorf scenario with Cuboid Splatting with prior with systematic error in the timeline of the prior (5h earlier release start). All metrics regarding relevance 1.

Prior	Resolu- tion	Reconstruc- tion quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
Brokdorf Prior2	30x30	60,98%	72,25%	(+11,27%P)	122,25%	(+61,26%P)
Brokdorf Prior2	10x10	31,8%	66,39%	(+34,59%P)	47,87%	(+16,07%P)
Brokdorf Prior2	100x100	127,51%	82,39%	(-45,12%P)	252,21%	(+124,7%P)

Can the method correct weather data errors from the prior scenario?

The following priors are used for evaluation, which contain the different types of weather data errors:

Emsland:

- Emsland local weather turbulences
- Emsland systematic wind speed error

Brokdorf:

- Brokdorf_Prior1
- Brokdorf local weather turbulences
- Brokdorf systematic wind speed error

The results of these experiments with Cuboid Splatting with prior are shown in Tables 17 and 18.

Table 17: Reconstruction results for the Emsland scenario with Cuboid Splatting using priors from priors with weather data errors from 25,000 paths with ADC. Metrics regarding relevance 1.

Prior	Resolution	Reconstruction quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
Emsland local turbulences	30x30	20,81%	19,57%	(-1,24%P)	40,14%	(+21,12%P)
Emsland syst. wind speed	30x30	24,83%	42,24%	(+17,4%P)	40,14%	(+15,3%P)
Emsland local turbulences	10x10	11,82%	24,99%	(+13,18%P)	17,74%	(+5,93%P)
Emsland syst. wind speed	10x10	8,43%	33,38%	(+24,95%P)	17,74%	(+9,31%P)
Emsland local turbulences	100x100	23,66%	11,14%	(-12,53%P)	145,31%	(+121,64%P)
Emsland syst. wind speed	100x100	41,27%	50,78%	(+9,52%P)	145,31%	(+104,04%P)

Table 18: Reconstruction results for the Brokdorf scenario with Cuboid Splatting using priors from priors with weather data errors from 25,000 paths with ADC. Metrics regarding relevance 1.

Prior	Resolution	Reconstruction quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
Brokdorf Prior1	30x30	8,91%	24,12%	(+15,21%P)	122,25%	(+113,33%P)
Brokdorf local turbulences	30x30	9,38%	22,08%	(+12,7%P)	122,25%	(+112,87%P)

Brokdorf syst. wind speed	30x30	13,79%	39,55%	(+25,76%P)	122,25%	(+108,46%P)
Brokdorf Prior1	10x10	8,79%	19,14%	(+10,35%P)	47,87%	(+39,08%P)
Brokdorf local turbulences	10x10	9,41%	21,55%	(+12,14%P)	47,87%	(+38,46%P)
Brokdorf syst. wind speed	10x10	7,07%	41,67%	(+34,6%P)	47,87%	(+40,8%P)
Brokdorf Prior1	100x100	20,01%	47,06%	(+27,05%P)	252,21%	(+232,2%P)
Brokdorf local turbulences	100x100	16,69%	13,23%	(-3,46%P)	252,21%	(+235,52%P)
Brokdorf syst. wind speed	100x100	18,39%	31,56%	(+13,17%P)	252,21%	(+233,82%P)

Can the method correct incorrectly (un)polluted parts of the prior scenario?

In both cases, priors were specifically constructed for each original scenario based on the prior scenarios EuRim-Emsland and Brokdorf_Prior1. These priors are used for evaluation:

Emsland:

- EuRim part of prior missing
- EuRim part of prior added

Brokdorf:

- Brokdorf Prior1 part of prior missing
- Brokdorf Prior1 part of prior added

An evaluation was performed to determine the extent to which the final mechanism of cuboid splatting with prior succeeds in adding cuboids and thus iodine load in parts of spacetime where the prior contains no load, and conversely, in removing this load from areas of spacetime incorrectly loaded in the prior during the correction step. The results regarding the reconstruction qualities of these experiments with cuboid splatting with prior are shown in Tables 19 and 20. Tables 21 and 22 also present an investigation of the sums of the load values of the affected space-time cells in the prior scenarios and the resulting reconstructions.

In the tables, the use of an unchanged prior scenario was compared for each scenario to be reconstructed with the use of the same prior scenario, but constructed in such a way that part of the prior scenario is missing or that an additional part has been inserted at a location that is actually unloaded in the prior and original scenarios.

Table 19: Reconstruction results for the Emsland scenario with cuboid splatting using prior from priors with parts missing/added from 25,000 paths with ADC. Metrics regarding relevance 1.

Prior	Resolu- tion	Reconstruc- tion quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
EuRim Prior	30x30	33,38%	80,37%	(+46,99%P)	40,14%	(+6,75%P)

EuRim Prior part missing	30x30	35,04%	80,54%	(+45,5%P)	40,14%	(+5,1%P)
EuRim Prior part added	30x30	30,72%	80,41%	(+49,69%P)	40,14%	(+9,42%P)
EuRim Prior	10x10	9,42%	79,81%	(+70,4%P)	17,74%	(+8,33%P)
EuRim Prior part missing	10x10	12,09%	80,54%	(+68,45%P)	17,74%	(+5,66%P)
EuRim Prior part added	10x10	11%	79,87%	(+68,87%P)	17,74%	(+6,74%P)
EuRim Prior	100x100	59,85%	91,95%	(+32,1%P)	145,31%	(+85,46%P)
EuRim Prior part missing	100x100	60,47%	92,11%	(+31,64%P)	145,31%	(+84,84%P)
EuRim Prior part added	100x100	57,56%	92,03%	(+34,46%P)	145,31%	(+87,74%P)

Table 20: Reconstruction results for the Brokdorf scenario with cuboid splatting using prior from priors with parts missing/added from 25,000 paths with ADC. Metrics regarding relevance 1.

Prior	Resolution	Reconstruction quality Scenario error [%]	Prior quality Scenario error [%] Baseline comparison 1 [%P]		Cuboid Splatting without prior Scenario error [%] Baseline comparison 2 [%P]	
Brokdorf Prior1	30x30	8,91%	24,12%	(+15,21%P)	122,25%	(+113,33%P)
Brokdorf Prior1 part missing	30x30	11,94%	24,38%	(+12,45%P)	122,25%	(+110,31%P)
Brokdorf Prior1 part added	30x30	9,66%	24,13%	(+14,48%P)	122,25%	(+112,59%P)
Brokdorf Prior1	10x10	8,79%	19,14%	(+10,35%P)	47,87%	(+39,08%P)
Brokdorf Prior1 part missing	10x10	7,8%	21,24%	(+13,44%P)	47,87%	(+40,07%P)
Brokdorf Prior1 part added	10x10	8,86%	19,16%	(+10,31%P)	47,87%	(+39,01%P)
Brokdorf Prior1	100x100	20,01%	47,06%	(+27,05%P)	252,21%	(+232,2%P)
Brokdorf Prior1 part missing	100x100	17,82%	47,16%	(+29,34%P)	252,21%	(+234,39%P)
Brokdorf Prior1 part added	100x100	20,11%	47,08%	(+26,96%P)	252,21%	(+232,1%P)

Table 21 contains the results of the reconstructions with cuboid splatting with prior based on the prior from which a part was removed. The sums of the load values in the affected part of the

space-time are given in each case. With regard to the prior scenario, these are the sums of the load values of the prior scenario in this area before the part was removed and afterwards (0,0). Then the sums of the load values in this part of space-time in the reconstruction based on the normal prior and based on the modified prior in which the part was missing. Finally, the sum of the load values in this part of space-time in the original scenario.

Table 21: Sums of the load values of the affected space-time cells in reconstruction with Cuboid Splatting with prior in comparison to unchanged prior scenario (normal) and prior with missing part (changed). (30x30, 25,000 paths with and without ADC)

	Prior scenario		Reconstruction		Original
	normal	changed	normal prior	changed prior	normal
Emsland rel. 0, with ADC	2238,84	0,0	34800,92	30799,54	36691,55
Emsland rel. 1, with ADC	662,81	0,0	16158,05	12735,73	14482,57
Emsland rel. 0, without ADC	2238,84	0,0	33239,18	30340,61	36691,55
Emsland rel. 1, without ADC	662,81	0,0	15926,23	13113,07	14482,57
Brokdorf rel. 0, with ADC	8419,05	0,0	26050,09	9444,22	11606,18
Brokdorf rel. 1, with ADC	2909,15	0,0	7911,20	7830,23	4725,86
Brokdorf rel. 0, without ADC	8419,05	0,0	23980,47	8070,05	11606,18
Brokdorf rel. 1, without ADC	2909,15	0,0	6974,91	6510,01	4725,86

Table 22 contains the results of the reconstructions with Cuboid Splatting with prior based on the prior in which an additional part was added. The sums of the load values in the affected part of the space-time are also specified here. With regard to the prior scenario, these are the sums of the load values of the prior scenario in this area before the part was added (0,0) and after. Then the sums of the load values in this part of space-time in the reconstruction based on the normal prior and based on the modified prior in which the part was added. Finally, the sum of the load values in this part of space-time in the original scenario (0,0).

Table 22: Sums of the load values of the affected space-time cells in reconstruction with Cuboid Splatting with prior in comparison to unchanged prior scenario (normal) and prior with added part (changed). (30x30, 25,000 paths with and without ADC)

	Prior scenario		Reconstruction		Original
	normal	changed	normal prior	changed prior	
Emsland rel. 0, with ADC	0,0	2238,84	0,0	1808,93	0,0
Emsland rel. 1, with ADC	0,0	294,96	0,0	41,49	0,0
Emsland rel. 0, without ADC	0,0	2238,84	0,0	1780,88	0,0
Emsland rel. 1, without ADC	0,0	294,96	0,0	43,49	0,0
Brokdorf rel. 0, with ADC	0,0	8419,05	0,0	1192,68	0,0
Brokdorf rel. 1, with ADC	0,0	3603,85	0,0	2,82	0,0
Brokdorf rel. 0, without ADC	0,0	8419,05	0,0	1451,11	0,0
Brokdorf rel. 1, without ADC	0,0	3603,85	0,0	36,32	0,0